

Tropical Proof Systems, Open Problem 1(a) – medium

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1 Preliminaries

Let

$$\mathbb{Q}_\infty := \mathbb{Q} \cup \{\infty\}, \quad a \oplus b := \min\{a, b\}, \quad a \odot b := a + b,$$

with $a \oplus \infty = a$ and $a \odot \infty = \infty$. A *tropical term* is an expression

$$c \odot x_1^{\odot d_1} \odot \cdots \odot x_n^{\odot d_n},$$

where $c \in \mathbb{Q}_\infty$ and $d_1, \dots, d_n \in \mathbb{N}_0$. A *tropical polynomial* is a finite tropical sum of tropical terms. A *min-plus inequality* is an inequality $f \leq g$ between tropical polynomials.

For every Boolean variable x , introduce two tropical variables x and $\bar{x}$, together with the equations

$$x \odot \bar{x} = 1, \quad x \oplus \bar{x} = 0.$$

Each equation is treated as the two opposite inequalities. A Boolean clause $\ell_1 \vee \cdots \vee \ell_k$ is encoded by

$$1 \leq \ell_1 \odot \cdots \odot \ell_k,$$

where the tropical variable corresponding to the literal $\neg x$ is $\bar{x}$.

An MP-PCR derivation is a sequence of min-plus inequalities. Every line is an encoded input clause, a Boolean axiom, an axiom $0 \leq 0$ or $p \leq \infty$, or follows from earlier lines by one of the rules

$$\frac{p \leq q \quad p' \leq q'}{p \oplus p' \leq q \oplus q'}, \quad \frac{p \leq q}{p \odot t \leq q \odot t}, \quad \frac{p \leq h \quad h \leq r}{p \leq r},$$

where t is a tropical term.

For tropical polynomials p, q , write $p \succ q$ if, for every monomial occurring in p , the coefficient of that monomial in p is strictly larger than its coefficient in q . An MP-PCR refutation ends with an inequality $p \leq q$ such that $p \succ q$ and q has a finite constant term. The size of a derivation is the total binary encoding length of all its lines.

A family $(F_n)_{n \geq 1}$ of CNF formulas is *explicit* if F_n can be constructed in time polynomial in n .

2 Problem formulation

Construct an explicit family $(F_n)_{n \geq 1}$ of unsatisfiable CNF formulas of size $n^{O(1)}$ such that every MP-PCR refutation of F_n has superpolynomial size:

$$\text{size}_{\text{MP-PCR}}(F_n) = n^{\omega(1)}.$$